

Damien MAÎTRE

MSc Engineering Student — CFD, HPC & Numerical Methods

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EDUCATION

- September 2026 **MSc Mobility & Physics engineering**
February 2027 *Student exchange at Politecnico di Milano, Italy*
International economy, transport operation, high performance computing, infrastructure finance.
- September 2023 **MSc Engineering — Fluid dynamics, Structural mechanics, Prototyping**
June 2027 *École nationale des ponts et chaussées (ENPC), Paris*
Structural mechanics and large-structure design, FEM applied to structural analysis, composite materials, metallurgy, complete fluid dynamics module, HPC, combustion.
 - **Tools:** OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, gmsh
 - **Projects:** Underwater sea-glider simulation and prototype ; closed-wing aircraft design and optimization ; wooden suspension bridge design, verification and construction.
- September 2021 **Bachelor — Classes préparatoires**
June 2023 *Lycée Kléber, Strasbourg*
Intensive programme in mathematics, physics and engineering sciences.

RESEARCH PROJECTS

- Complete CFD developer-first suite**
Personal project — OpenFOAM, Gmsh (forked), Python, C++
Open and user-friendly aerodynamical loads studies, tailored for very high Re axisymmetric problems.
- Structural FEM & Eurocode Verification suite**
Personal project — OpenSeesPy, Python
End to end structure design, visualization and verification tools.

PROFESSIONAL EXPERIENCE

- February 2026 **Rail infrastructure engineering internship — TRACTEBEL FRANCE**
July 2026 *CDG Express project, Saint-Ouen*
 - Construction compliance verification for the CDG Express, a 2.5B€ high-speed rail project,
 - Structural analysis and Eurocode verifications of civil engineering structures,
 - Review of structural calculation reports and as-built site surveys,
 - Drafting technical reports and recommendations for the client.
- July 2025 **Digital Twin PM intern — SNCF Voyageurs**
January 2026 *Industry & digital solutions, Saint-Denis*
 - Management of a 150k€ digital project within the rolling stock maintenance industry,
 - Coordination of 15 3D artists across the country's maintenance plants,
 - Project ownership and reporting to internal clients (plant directors, Industrial Director, etc.),
 - End-to-end procurement and tendering process.

TECHNICAL SKILLS

- CFD & FEM** OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, Gmsh
Programming 10+ years of experience with Python, C++ and UNIX environments

LANGUAGES

French — Native · **English** — Proficient (C1) · **Italian** — Working proficiency (B2) · **German** — Working proficiency (B2)

ACTIVITIES & COMMUNITY INVOLVEMENT

- Rescuer — French Red Cross**
Volunteer first-aid responder
First-aid responder at public events (marathons, festivals, Paris-Plage) and on-call with Paris Fire Brigade.
- Director — Student Musical Ensemble**
BridgeWay Comédie Musicale, Champs-Sur-Marne
Directed an ensemble of 75 artists and a creative/technical leadership team of 12 with 4.5k€ budget.
- General Secretary — Student Association**
Union des Elèves de l'Ecole Nationale des Ponts et Chaussées
Management of a team of 15 for professional and social events with 1,000+ attendees, Administrative management of a registered French non-profit (Loi 1901).